

Mila Victório Pessotto

Agronomist | Applied Research, Sustainable Production Systems and Soil Health

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SUMMARY

Agronomist with a Ph.D. and international experience in applied research focused on sustainable cropping systems. Experienced in designing, conducting, and analyzing field and laboratory experiments in soil science, soil fertility and soil health, plant nutrition, cover crops, and nitrogen management in row crops. Skilled in experimental design, statistical analysis, data science, technical communication, and collaborating with universities, public institutions, farmers, and crop consultants to translate agronomic data into practical recommendations.

CORE EXPERTISE

Soil Health | Soil Fertility | Sustainable Cropping Systems | Experimental Design and Data Analysis | Statistical Analysis using R | Written and Oral Scientific Communication | Extension and Outreach | Field and Laboratory Research | Mentoring | Cover Crops | Crop Nutrition | Nitrogen Management | Data Management and Visualization

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate at Iowa State University

2025 – 2026

- Conducted and analyzed soybean response-to-climate-change experiments, focusing on irrigation and rainfall variability in Iowa, USA.
- Mentored undergraduate and graduate students in experimental design, data analysis, and scientific presentations.
- Performed laboratory analyses including nitrate, ammonium, soil enzymes, and soil organic carbon and nitrogen.
- Authored and published peer-reviewed scientific papers and extension publications.
- Organized and presented field days for farmers, students, faculty, and agricultural consultants.

Senior Research Assistant (Ph.D.) at Iowa State University

2022 – 2025

- Designed, conducted, and analyzed research on crop rotation, corn productivity, no-till systems, corn nitrogen response, corn-cereal rye rotations, and nitrogen cycling in Iowa.
- Conducted laboratory analyses to evaluate microbial activity, plant-available nitrogen, soil quality, and soil stability.
- Developed and evaluated soil health assessment methods.
- Collaborated on multidisciplinary projects involving cropping systems, crop rotation, and agricultural sustainability.
- Authored peer-reviewed scientific publications and extension articles for farmers and crop consultants.
- Supervised undergraduate students in laboratory research and undergraduate thesis projects.

Research Assistant (M.S.) at University of Arkansas

2019 – 2022

- Planned, conducted, and analyzed soil fertility experiments focusing on nitrogen management, cover crops, and plant physiology.
- Designed and conducted controlled-environment growth chamber experiments to determine cardinal temperatures for eight cover crop species.
- Collaborated on multiple soil fertility projects in Arkansas, including rice response to nitrogen fertilization, soybean response to phosphorus and potassium fertilization, and corn response to nitrogen, magnesium, and sulfur fertilization.

EDUCATION

Ph.D., Crop Production and Physiology <i>Iowa State University (ISU)</i>	2025
M.S., Crop, Soil, and Environmental Sciences <i>University of Arkansas (UARK)</i>	2022
B.S., Agricultural Engineering <i>College of Agriculture "Luiz de Queiroz" / University of São Paulo (USP)</i>	2018

SELECTED PUBLICATIONS

A complete publication list is available at: mvpessotto.github.io/

Pessotto, M.V., Licht, M.A., Adjuik, T.A., Marcos, F.M., Matthiesen, R., O'Brien, P.L., Robertson, A. & McDaniel, M.D. 2026. Hillslope position affects soil nitrogen dynamics and maize fertilizer needs following a cereal rye cover crop. *Field Crop Research Journal*. Volume 344, 110516, ISSN0378-4290.
<https://doi.org/10.1016/j.fcr.2026.110516>.

Pessotto, M.V., McDaniel, M.D., Larson, A.C. & Licht, M.A. 2026. Robel Pole Method Can Estimate Cereal Rye Biomass to Optimize Termination Timing. *Crop, Forage & Turfgrass Management*, 12, e70152.
<https://doi.org/10.1002/cft2.70152>

Pessotto, M.V., Roberts, T.L., dos Santos, C.L., Hoegenauer, K., Bertucci, M., Ross, J. & Savin, M. 2025. Use of Growing Degree Days to Predict Aboveground Biomass and Total Nitrogen Accumulation of Winter Cover Crops. *Agrosystems, Geosciences & Environment*. V.8, issue 4. <https://doi.org/10.1002/agg2.70249>

Pessotto, M.V., Roberts, T.L., Bertucci, M., dos Santos, C.L., Ross, J. & Savin, M. 2023. Determining Cardinal Temperatures for Eight Cover Crop Species. *Agrosystems, Geosciences & Environment*. V.6, e20393.
<https://doi.org/10.1002/agg2.20393>

LEADERSHIP & HONORS

- Author of technical guides for 34 cover crop species in collaboration with USDA-NRCS (2024).
- Outstanding Paper Award - *Agrosystems, Geosciences & Environment* (2025).
- Outstanding Ph.D. Student Award - Iowa State University Department of Agronomy (2025).
- Agronomy Teaching Fellowship - Iowa State University Department of Agronomy (2023 and 2024).
- First Place Poster Awards at the Midwest Cover Crop Council & South Dakota Soil Health Coalition (2023) and the ASA/CSSA/SSSA Annual Meeting (2022).
- Organized and presented field days and training programs on soil health and cover crops for farmers, consultants, students, and partner organizations.
- Teaching experience in agronomy and soil fertility courses and laboratories; mentored undergraduate and graduate students.
- Ad hoc reviewer for *Crop, Forage & Turfgrass Management* and active participant in graduate student committees.

TECHNICAL TOOLS & LANGUAGES

Methods and tools: R for statistical analysis and visualization, experimental design, field-trial and laboratory protocols, soil and plant analyses, data management

Portuguese – Native | English – Fluent | Spanish - Intermediate